IE540
L5. Temporal Difference Methods

KAIST ISE
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Overview

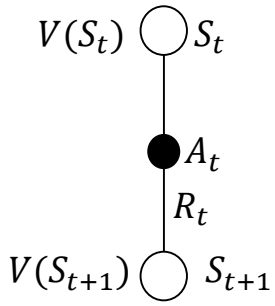
- Model is known → DP (in Lecture 3)
  - Evaluation (prediction) : Iterative policy evaluation
  - Control (finding optimal solution)
    - PI → GPI → VI
    - Async. DP
- Model is unknown → RL (by use of interaction samples)
  - Value based RL
    - MC method : Evaluation, Control (in Lecture 4)
    - TD method : Q, SARSA (in Lecture 5)
    - n-step bootstrapping (in Lecture 6)
    - value function approximation (in Lecture 7)
    - NN, DQN (Lecture 8)
  - Policy based RL (Lecture 9~11)
    - Derivative free methods
    - Policy gradient

Temporal difference learning

Learning x from y at the rate α

$$x \overset{\alpha}{\leftarrow} y \equiv x \leftarrow x + \alpha(y - x)$$

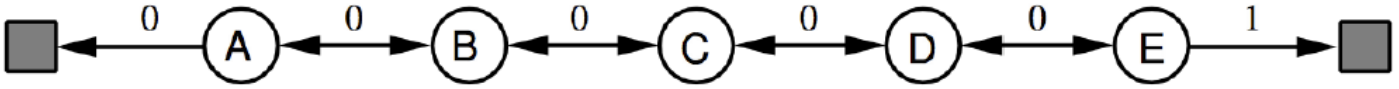
- MC PE
  - $\tau = (S_0, A_0, R_0, S_1, A_1, R_1, \dots, S_T)$
  - $V(S_t) \leftarrow V(S_t) + \alpha_t\big(G_t - V(S_t)\big) \equiv V(S_t) \overset{\alpha_t}{\leftarrow} G_t$
    - update using actual return G_t (MC target)
- TD(0) (Simple Temporal Difference) PE
  - recall Bellman exp. eq. : $V^\pi(s) = \mathbb{E}_{\substack{a\sim\pi(a|s) \\ r,s'\sim p(r,s'|s,a)}} [r + \gamma V^\pi(s')]$
  - Update $V(S_t)$ toward estimated return $R_t + \gamma V(S_{t+1})$
  - $V(S_t) \leftarrow V(S_t) + \alpha_t\big(R_t + \gamma V(S_{t+1}) - V(S_t)\big) \equiv V(S_t) \overset{\alpha_t}{\leftarrow} R_t + \gamma V(S_{t+1})$
    - TD target : $R_t + \gamma V(S_{t+1})$
    - TD error : $\delta_t = R_t + \gamma V(S_{t+1}) - V(S_t)$



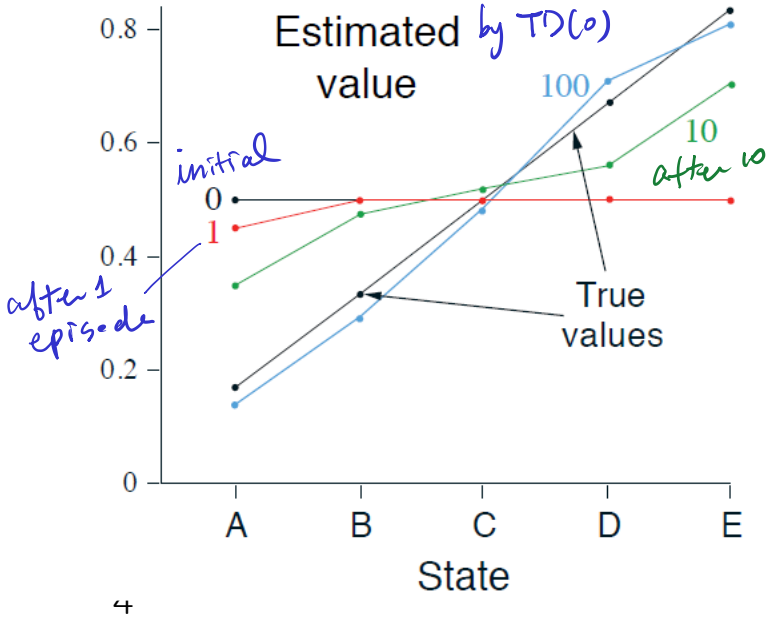
➢ **Algorithm TD(0) PE**

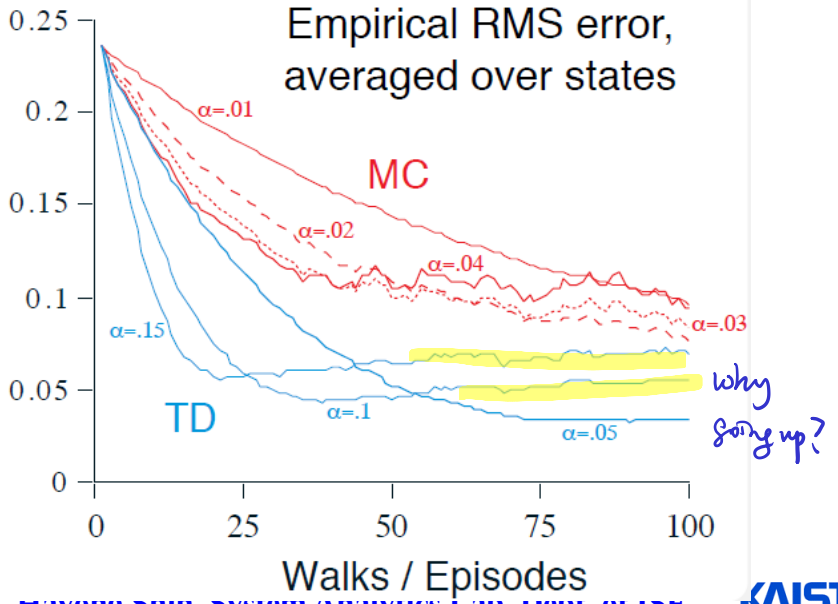
- initialize $V(s)$ to any value for $\forall s$ and $V(terminal) = 0$
- for $k = 1..n$ (n : # of episodes) (or until convergence)
  - initialize S_0
  - for each step $t = 0,1, \dots$ (until S_t is $terminal$)
    - Choose action $A_t \sim \pi(.|S_t)$
    - take action A_t, observe R_t, S_{t+1}
    - $V(S_t) \leftarrow V(S_t) + \alpha_t(R_t + \gamma V(S_{t+1}) - V(S_t))$

Example : Random walk MRP (RLAI ex.6.2)

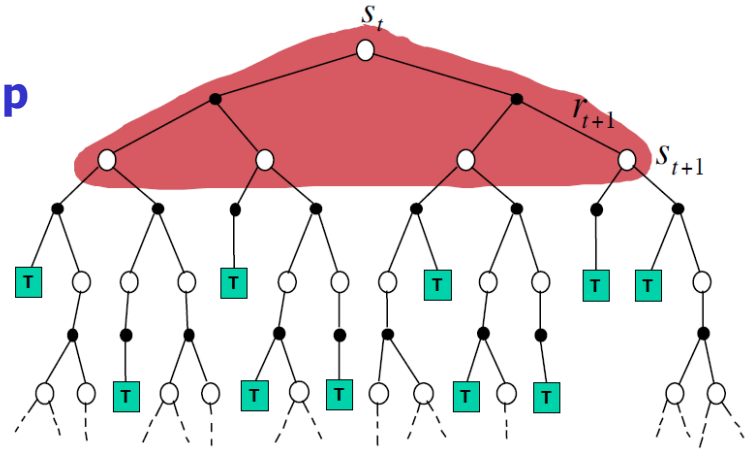


- State S = {A,B,C,D,E,■}, Transition : left or right equally
- Reward : 1 for moving from E to ■, 0 otherwise, $\gamma = 1$
- True value $\left(V(A), V(B), V(C), V(D), V(E)\right)$ =
- Comparing MC vs TD(0)
  - initial $V(*) = 0.5$, learning rate α : constant
  - TD(0) converges faster
  - constant α hinders convergence below some limit

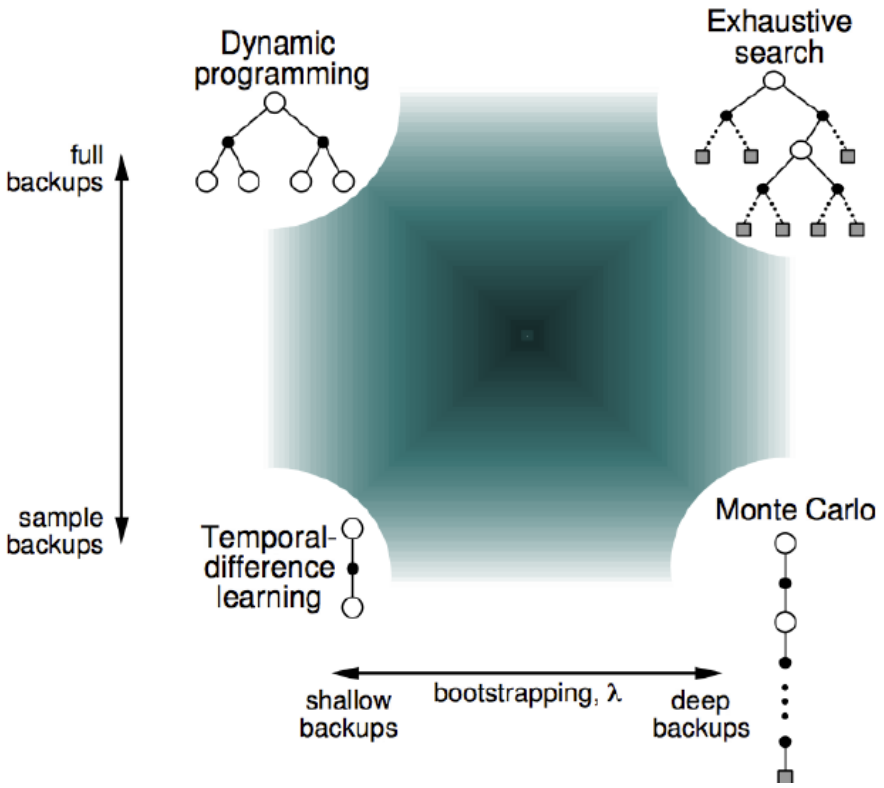




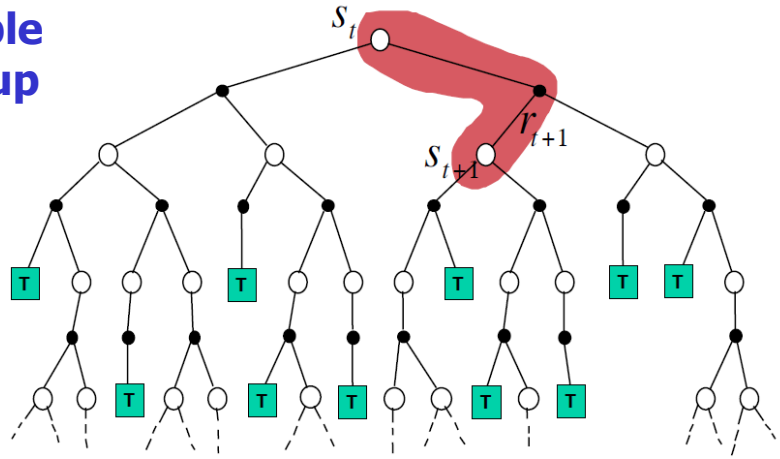
Full backup



DP Backup : $V(S_t) = \mathbb{E}_\pi[R_t + \gamma V(S_{t+1})]$

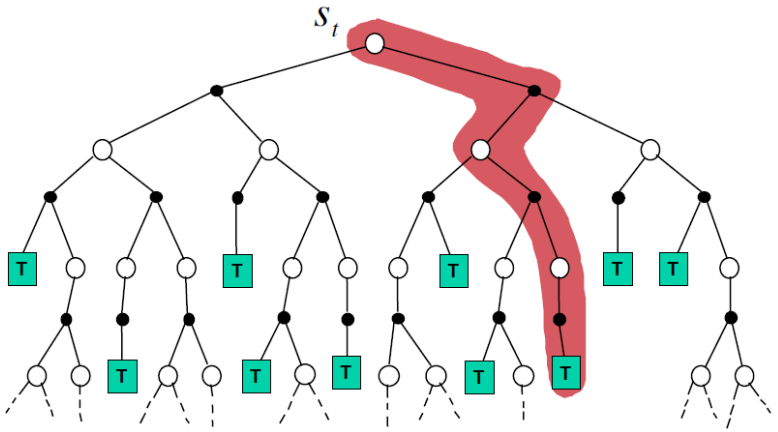


Sample backup



TD Backup

$$V(S_t) \leftarrow V(S_t) + \alpha_t\big(R_t + \gamma V(S_{t+1}) - V(S_t)\big)$$



MC Backup

$$V(S_t) \leftarrow V(S_t) + \alpha_t\big(G_t - V(S_t)\big)$$

MC vs TD

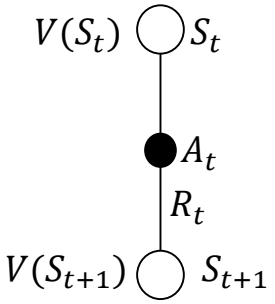
- MC
  - learns from complete episode
    - episodic tasks only
  - using return G_t as target for $V(S_t)$
    - no bootstrapping
    - no bias
      - G_t is an unbiased estimator
      - insensitive to initial $V(.)$
    - variance :
      - return ≈ sum of many rewards
    - converges to $V^{\pi}(s)$
      - suffer from slow convergence
      - mainly due to high var

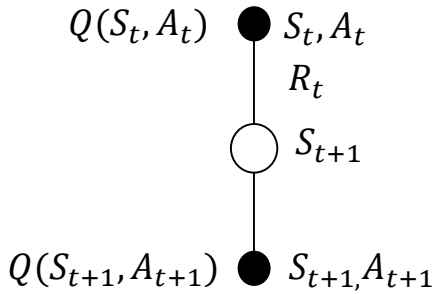
- TD
  - learns from incomplete sequence
    - episodic or continuing tasks
  - using return $R_t + \gamma V(S_{t+1})$ as target for $V(S_t)$
    - bootstrapping
      - DP also bootstraps
    - some bias
      - $R_t + \gamma V^{\pi}(S_{t+1})$ is unbiased
      - but $R_t + \gamma V(S_{t+1})$ is.
        - » because $V(.)$ arbitrarily initialized
        - » sensitive to initial $V(.)$
        - » need to set $V(terminal) = 0$
    - variance
    - converges to $V^{\pi}(s)$
      - bias gets smaller.
        - »

- TAQ. Commonality of MC &TD?
  -
  -
- TAQ. Which one is faster, MC or TD?
  -
  -

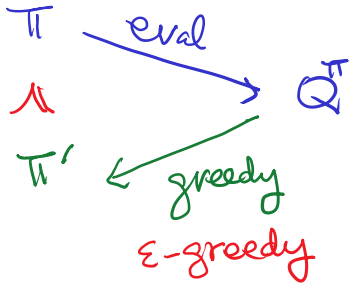
TAQs on TD(0) PE

- TD(0) (Simple Temporal Difference) PE
  - Bellman exp. eq. : $V^{\pi}(S_t) = \mathbb{E}_{\substack{A_t \sim \pi(a|S_t) \\ R_t, S_{t+1} \sim p(r,s'|S_t,A_t)}} [R_t + \gamma V^{\pi}(S_{t+1})]$
    - $V(S_t) \approx R_t + \gamma V(S_{t+1})$
  - Update $V(S_t)$ toward estimated return: $V(S_t) \overset{\alpha_t}{\leftarrow} R_t + \gamma V(S_{t+1})$
    - $V(S_t) \leftarrow V(S_t) + \alpha_t \big(R_t + \gamma V(S_{t+1}) - V(S_t)\big)$
  - Why is it called TD(0)? Soon, we will cover TD(1) ≈ MC , TD(λ)
  - How can we learn action value?
    - $Q(S_t, A_t) \overset{\alpha_t}{\leftarrow} R_t + \gamma Q(S_{t+1}, A_{t+1})$: TD(0) PE, Q version
    - TAQ. How to select A_{t+1}? $A_{t+1} \sim$ [ ]
  - TAQ. Why update $V(S_t)$ instead of $V(S_{t+1})$?
    - In MC, we use $V(S_t) \leftarrow V(S_t) + \alpha_t \big(G_t - V(S_t)\big)$
      - In TD(0), $R_t + \gamma V(S_{t+1})$ is used in place of G_t
    - Usually $V(S_{t+1})$ is closer to the terminal state
      - $V(S_{t+1})$ is more certain, usually
    - If estimation uncertainty of $V(s)$ is known,
      - we can determine which side to update
      - $V(S_t)$ or $V(S_{t+1})$ or both





ϵ-greedy policy improvement



- greedy policy improvement (from DP lecture) :
  - π : any policy
  - Q^π : action value of π
  - π' = greedy (Q^π) → $\pi' \geqslant \pi$ (i.e. $V^{\pi'}(s) \geq V^\pi(s)$)
- ϵ-greedy policy improvement :
  - π : any policy w/ $\pi(a|s) \geq \frac{\epsilon}{m}$ for $\forall(s,a)$, where $m = |A|$
    - e.g. $\pi = \epsilon-$greedy (Q) for some Q $\quad (Q \neq Q^\pi)$
  - Q^π : action value of π
  - π' = ϵ-greedy (Q^π) → $\pi' \geqslant \pi$ (i.e. $V^{\pi'}(s) \geq V^\pi(s)$)

$$\sum_a \pi'(a|s)\, Q^\pi(s,a) = \sum_a \frac{\epsilon}{m} Q^\pi(s,a) + (1-\epsilon) \max_a Q^\pi(s,a)$$

$$\geq \sum_a \frac{\epsilon}{m} Q^\pi(s,a) + (1-\epsilon) \sum_a w_a Q^\pi(s,a)\,, \text{where} \sum_a w_a = 1, w_a \geq 0$$

$$= \sum_a \frac{\epsilon}{m} Q^\pi(s,a) + (1-\epsilon) \sum_a \frac{\pi(a|s) - \epsilon/m}{1-\epsilon} Q^\pi(s,a) \quad \text{by choosing } w_a = \frac{\pi(a|s) - \epsilon/m}{1-\epsilon}$$

$$= \sum_a \pi(a|s) Q^\pi(s,a) = V^\pi(s)$$

Selecting the first action w/ π' is better than using π.

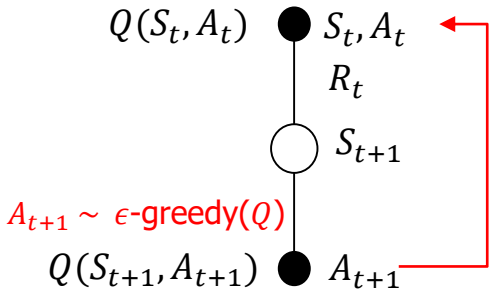
This argument can be repeated to 2nd action, 3rd action, …→ $\pi' \geqslant \pi$

SARSA : On-policy TD control

- Similar to On-policy MC Control using GLIE policy
- Use TD target $R_t + \gamma Q(S_{t+1}, A_{t+1})$, where $A_{t+1} \sim \epsilon$-greedy(Q) at S_{t+1}

➢ **Algorithm SARSA (On-policy control)**

- initialize $Q(s,a)$ to any value for $\forall(s,a)$ and $Q(terminal, a) = 0$
- for $k = 1..n$ (n : # of episodes) (or until convergence)
  - initialize S_0
  - choose action $A_0 \sim \epsilon$−greedy(Q) at S_0
  - for each step $t = 0,1,\ldots$ (until S_t is $terminal$)
    - take action A_t, observe R_t, S_{t+1}
    - Choose action $A_{t+1} \sim \epsilon$−greedy(Q) at S_{t+1}
    - $Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\big(R_t + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\big)$

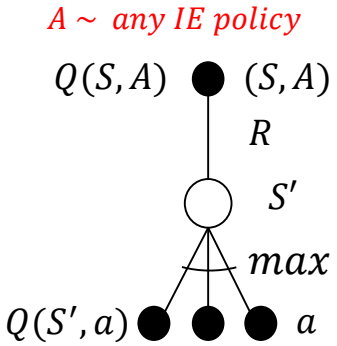


➢ **Algorithm SARSA (On-policy control) – w/o time index** t

- initialize $Q(s,a)$ to any value for $\forall(s,a)$ and $Q(terminal, *) = 0$
- for $k = 1..n$ (n : # of episodes) (or until convergence)
  - initialize S
  - choose action $A \sim \epsilon$−greedy(Q) at S
  - for each step (until S is $terminal$)
    - take action A, observe R, S'
    - Choose action $A' \sim \epsilon$−greedy(Q) at S'
    - $Q(S,A) \leftarrow Q(S,A) + \alpha\big(R + \gamma Q(S', A') - Q(S,A)\big)$
    - $S \leftarrow S',\ A \leftarrow A'$

Q-Learning : Off-policy TD control

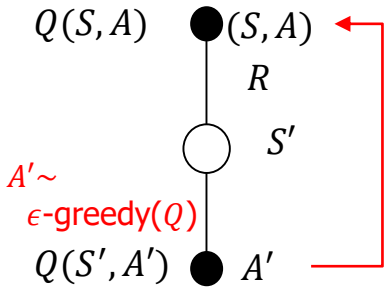
- Q-learning (Watkins, 1989)
  - Bellman Opt. Eq. $Q^*(S,A) = R(S,A) + \gamma \mathbb{E}_{S' \sim p^A_{S,S'}} \left[\max_a Q^*(S,a) \right]$
  - Q value update $Q(S,A) \overset{\alpha}{\leftarrow} R + \gamma \max_a Q(S',a)$
    - $x \overset{\alpha}{\leftarrow} y \equiv x \leftarrow x + \alpha(y - x)$

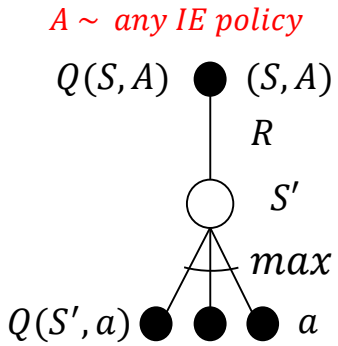


➢ **Algorithm Q-Learning (Off-policy control)**

- initialize $Q(s,a)$ to any value for $\forall(s,a)$ and $Q(terminal,*) = 0$
- for $k = 1..n$ (n : # of episodes) (or until convergence)
  - initialize S
  - for each step (until S is $terminal$)
    - Choose action $A \sim \mu(.|S)$: behavior policy, e.g. $\epsilon-$greedy(Q)
    - take action A, observe R, S'
    - $Q(S,A) \leftarrow Q(S,A) + \alpha \left(R + \gamma \max_a Q(S',a) - Q(S,A) \right)$
    - $S \leftarrow S'$

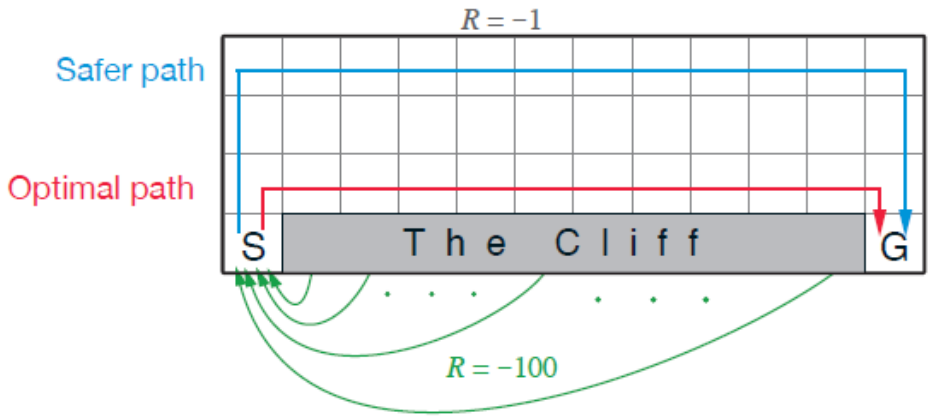
Q-Learning



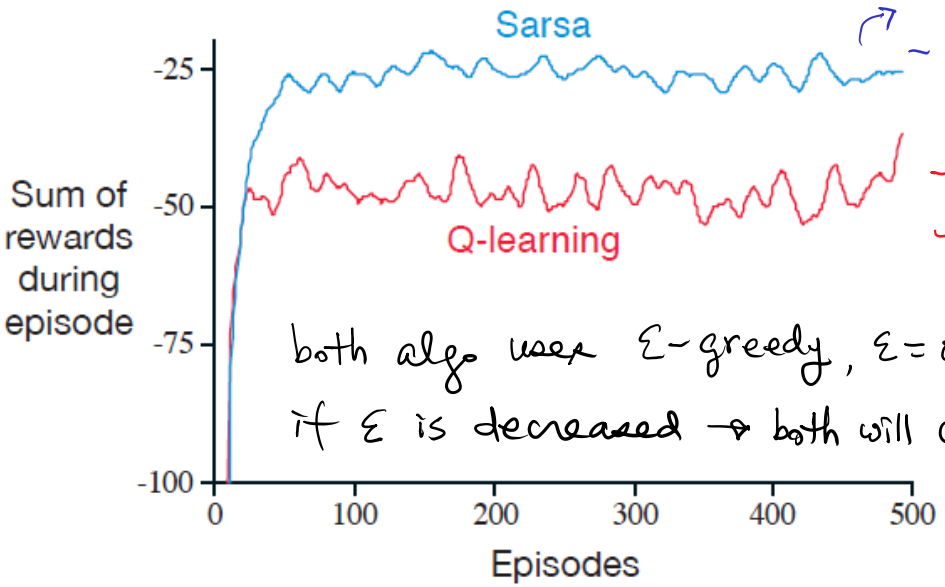


- Comparing SARSA vs Q-learning
  - SARSA $Q(S,A) \overset{\alpha}{\leftarrow} R + \gamma Q(S',A')$
    - $A' \sim \epsilon$-greedy(Q) at S'
    - A' is used for target and behavior →
  - Q-learning $Q(S,A) \overset{\alpha}{\leftarrow} R + \gamma \max_a Q(S',a)$
    - update target is not dependent on μ →
    - TAQ. Why isn't importance weight used here?
  - TAQ. Which algorithm converges faster to the optimal?
    -
- Behavior policy μ in Q-learning
  - derived from Q → for exploitation
  - usually $\epsilon-$greedy(Q) is used
  - TAQ. Does μ need to be GL?
    - GL (Greedy in the Limit): $\mu \to$ greedy(Q)
  - TAQ. Does μ need to be IE?
    - IE (Infinite Exploration) : $N(s,a) \to \infty$ for $\forall(s,a)$

Cliff walking example (RLAL ex.6.6)



$r = \begin{cases} -1 & \text{except cliff} \Rightarrow \text{shortest path } S \to G \\ -100 & \text{if cliff} \end{cases}$



- learns safer path
- online performance is better

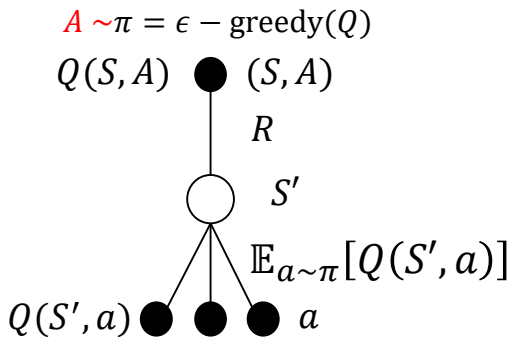
- learns optimal path
- falls off the cliff because of $\varepsilon = 0.1$
  → worse online performance

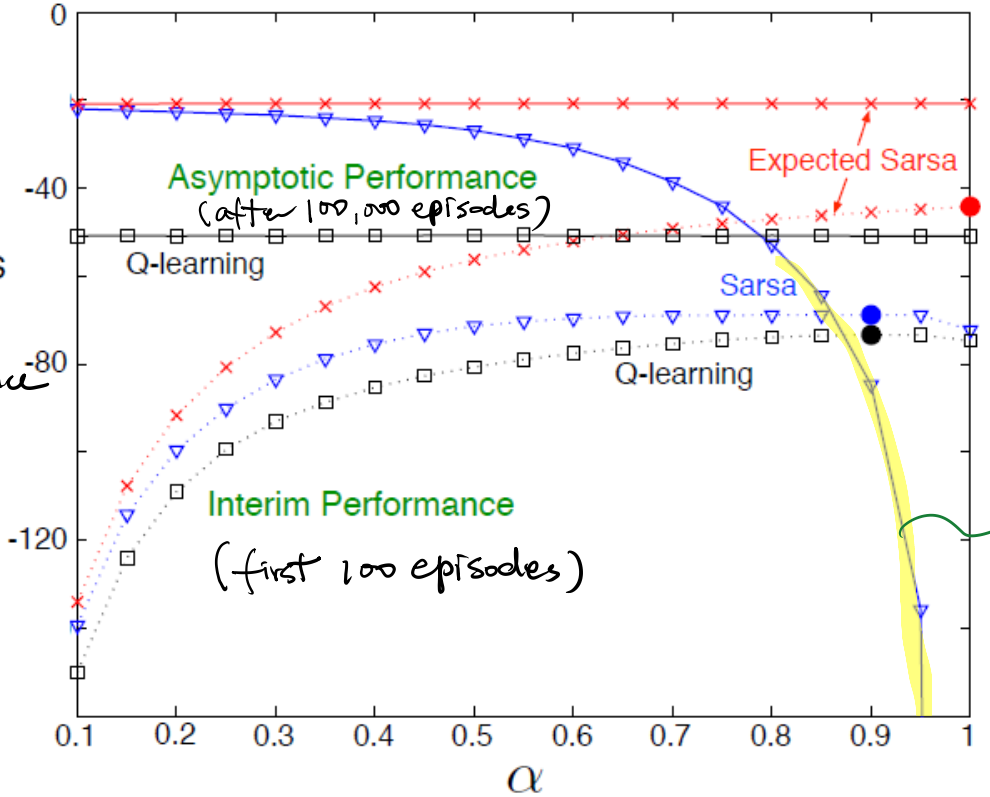
both algo uses ε-greedy, $\varepsilon = 0.1$

if ε is decreased → both will converge to optimal path

Expected SARSA

- SARSA $Q(S,A) \overset{\alpha}{\leftarrow} R + \gamma Q(S', A')$
- Q-learning $Q(S,A) \overset{\alpha}{\leftarrow} R + \gamma \max_a Q(S', a)$
- Expected SARSA $Q(S,A) \overset{\alpha}{\leftarrow} R + \gamma \mathbb{E}_{a\sim\pi}[Q(S', a)]$
  - on-policy → online performance is better
    - off-policy expected SARSA is also possible : $A\sim\mu$ & $\mathbb{E}_{a\sim\pi}[Q(S', a)]$
      - » $\mu = \epsilon - \text{greedy}(Q)$ & $\pi = \text{greedy}(Q)$ ➔
  - update target is much more stable than SARSA



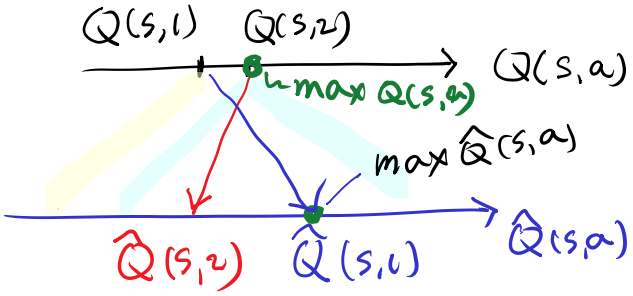


- Exp. SARSA shows
  - the best asymptotic online performance (when ε is const.)
  - the best interim online performance in wide range of α

As $\alpha \to 1$, SARSA suffers from $\text{Var}_{A'\sim\pi}[Q(S',A')]$

Maximization bias

- Maximization is used for control : $\max_a Q(S', a)$
  - SARSA: $\epsilon - \text{greedy}(Q)$ uses $\arg\max_a Q(S, a)$ to select action
  - Q-learning: $Q(S, A) \overset{\alpha}{\leftarrow} R + \gamma \max_a Q(S', a)$
- Maximization bias
  - Let $\hat{Q}(s, a)$ is an unbiased estimator of $Q(s, a)$, obtained by samples
    - $\mathbb{E}[\hat{Q}(s, a)] = Q(s, a)$
  - $\max_a \hat{Q}(s, a)$ is often used for $\max_a Q(s, a)$ in control
    - Is it unbiased? No
  - $\mathbb{E}\left[\max_a \hat{Q}(s, a)\right] \geq \max_a \mathbb{E}[\hat{Q}(s, a)] = \max_a Q(s, a)$
    - [ ] inequality : $\mathbb{E}[\varphi(f(X))] \geq \varphi(\mathbb{E}[f(X)])$ if $\varphi(.)$ is convex
    - [ ] is a convex function
  - $\max_a \hat{Q}(s, a)$ is positively biased from $\max_a Q(s, a)$
    - larger [ ] ➔ larger bias
    - more sample for $\hat{Q}(s, a)$ ➔ less bias



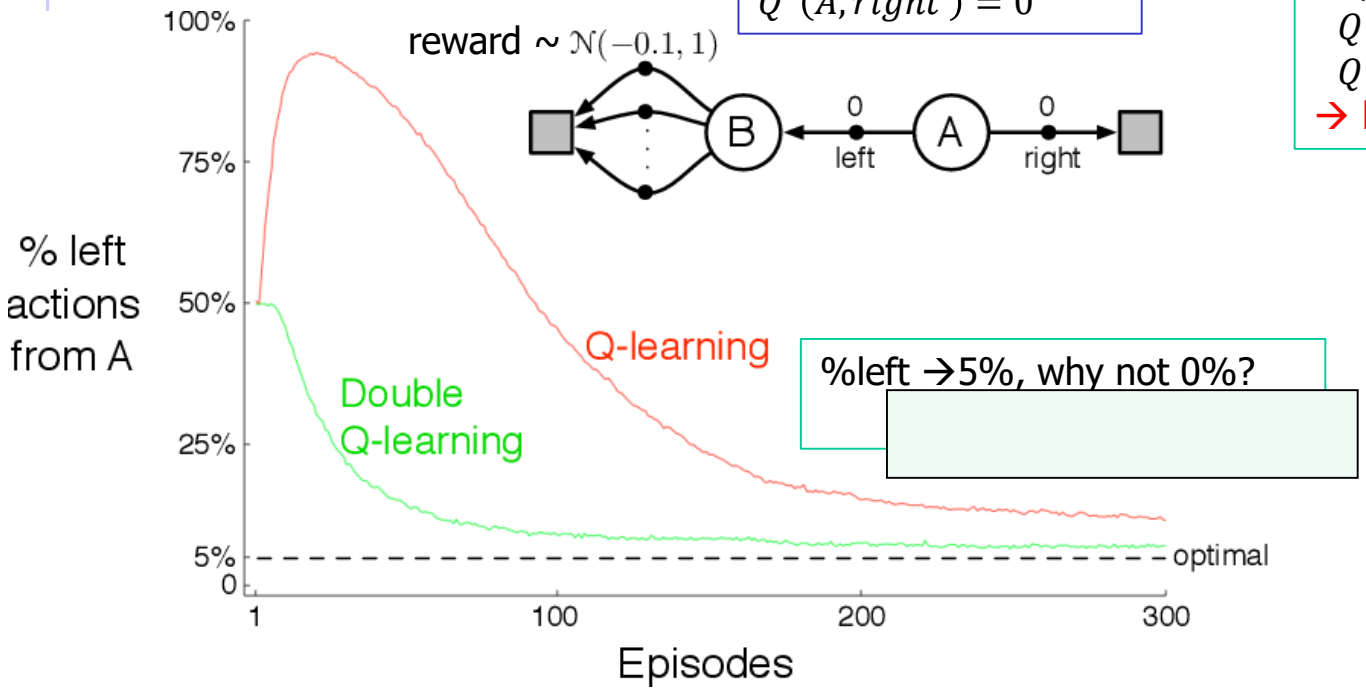
Maximization bias (RLAI Ex.6.7)

action $i \in \mathcal{A}(B) = \{1,2,..,n\}$
reward $R(B,i) = -0.1$

$Q^*(B,i) = -0.1$ for $\forall i$
$V^*(B) = -0.1$
$Q^*(A,left) = -0.1$
$Q^*(A,right) = 0$

Set 1
$r(B,1)$ = -0.1
$r(B,2)$ = +0.3
$r(B,3)$ = -0.2
→ max $Q(B,i)$ = +0.3
$Q(B,1)$ = -0.1
$Q(B,2)$ = +0.3
$Q(B,3)$ = -0.2

Using estimation set 1
$V(B)$= max $Q(B,i)$ = +0.3
$Q(A,left)$ = +0.3
$Q(A,right)$ = 0
→ left is taken



%left →5%, why not 0%?

Set 2
$r(B,1)$ = +0.2
$r(B,2)$ = -0.3
$r(B,3)$ = -0.1
→ max $Q'(B,i)$ = +0.2
$Q'(B,1)$ = +0.2
$Q'(B,2)$ = -0.3
$Q'(B,3)$ = -0.1

Using double Q
i^* =argmax $Q'(B,i)$=1
$V(B) = Q(B,i^*)$ = -0.1

$Q(A,left)$ = -0.1
$Q(A,right)$ = 0
→ right is taken

If Q'(B,i*) is big since i* is good, it will be reflected in Q.

If Q'(B,i*) is big by random, Q(B,i*) won't be big.

Double Q learning (Hasselt, AAAI 2010)

Single Q-learning

$$Q(S,A) \overset{\alpha}{\leftarrow} R + \gamma \max_a Q(S',a)$$

$$\equiv \; Q(S,A) \overset{\alpha}{\leftarrow} R + \gamma Q\left(S', \arg\max_a Q(S',a)\right)$$

Double Q-learning

select a from Q^A

$$Q^A(S,A) \overset{\alpha}{\leftarrow} R + \gamma Q^B\left(S', \arg\max_a Q^A(S',a)\right)$$

get value from Q^B

- **Algorithm Double Q-Learning**
- initialize $Q^A(s,a)$ and $Q^B(s,a)$ to any value for $\forall(s,a)$ and $Q^{A\ or\ B}(terminal,*) = 0$
- for $k = 1..n$ (n : # of episodes) (or until convergence)
  - initialize S
  - for each step (until S is $terminal$)
    - Choose action $A \sim \mu(.|S)$: ϵ−greedy$(Q^A + Q^B)$
    - take action A, observe R, S'
    - w.p. 0.5, update Q^A : $Q^A(S,A) \overset{\alpha}{\leftarrow} R + \gamma Q^B\left(S', \arg\max_a Q^A(S',a)\right)$
    - otherwise, update Q^B : $Q^B(S,A) \overset{\alpha}{\leftarrow} R + \gamma Q^A\left(S', \arg\max_a Q^B(S',a)\right)$
    - $S \leftarrow S'$
- return $Q =$

TAQ. How about this update?

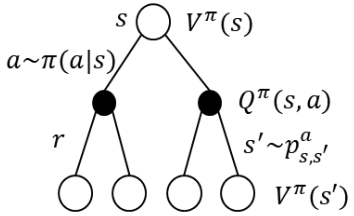
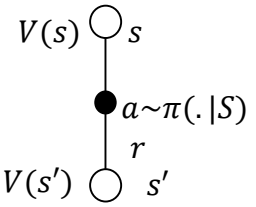
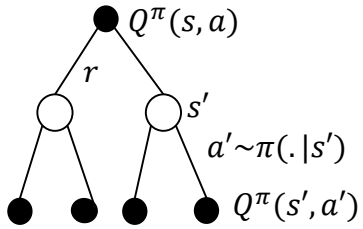
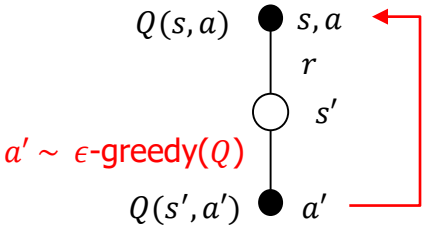
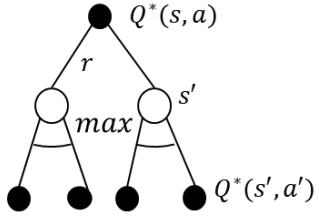
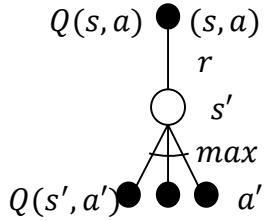
$$Q^A(S,A) \overset{\alpha}{\leftarrow} R + \gamma Q^A\left(S', \arg\max_a Q^B(S',a)\right)$$

TAQ. After maximization bias is gone, is double Q-learning wasting experiences?

Summary of TD(0)

- Introduced model free TD(0) methods
  - TD(0) : one-step temporal difference
  - sample backup
  - using Bellman equation
    - Markov state → convergence
  - bootstrap → biased
    - consistent : bias decreases
  - smaller variance of target
    - TD(0) methods : TD target $R_t + \gamma V(S_{t+1})$
    - MC method : discounted sum of rewards as target
- Policy evaluation : TD(0) PE
- Control (using GPI form)
  - on-policy : SARSA, Expected SARSA
  - off-policy : Q-learning, , Expected SARSA
- Maximization bias : over-estimation of maximum
  - avoided by Double Q-learning

Summary : DP vs TD

	Full backup (DP)	Sample backup (TD)
Bellman Expectation Eq. for $V^{\pi}(s)$	 Iterative policy evaluation $V^{\pi}(s) \leftarrow \mathbb{E}[r+\gamma V^{\pi}(s')\|s]$	 TD(0) PE $V^{\pi}(s) \overset{\alpha}{\leftarrow} r+\gamma V^{\pi}(s')$
Bellman Expectation Eq. for $Q^{\pi}(s,a)$	 Policy iteration (Q) $Q(s,a) \leftarrow \mathbb{E}[r+\gamma Q(s',a')\|s,a]$	 SARSA $Q(s,a) \overset{\alpha}{\leftarrow} r+\gamma Q(s',a')$
Bellman Optimality Eq. for $Q^{*}(s,a)$	 Value iteration (Q) $Q(s,a) \leftarrow \mathbb{E}\left[r+\gamma \max_{a'} Q(s',a')\|s,a\right]$	 Q-learning $Q(s,a) \overset{\alpha}{\leftarrow} r+\gamma \max_{a'} Q(s',a')$